

BOREAS Hyperparameter Configuration

Configuration used for the updated model with encoder width $d_{\text{model}} = 64$.

Hyperparameter	Value
<i>Model architecture</i>	
Ensemble members	5
HETI graph representation width	64
Exogenous representation width	64
HETI graph-network layers	2
Attention heads	4
RTCU/GRU hidden state width	256
Transition-head hidden width	256
Transition-head depth	1 layer
Inverse-head hidden width	256
Total forward parameters	approximately 3.619M
<i>Training objective</i>	
Main objective	One-step observation-increment Gaussian NLL
Log-variance bound penalty weight	0.01
Inverse-loss weight	0.03
Inverse magnitude-loss weight	1.0
Inverse positive-sample weight exponent	1.0
Inverse-loss warm-up	1,000 updates
Inverse-loss components	Prevalence-weighted BCE for nonzero-action detection and normalized Smooth L1 for nonzero-action magnitude
Multi-step rollout loss	Not used in final training
<i>Optimization</i>	
Optimizer	Adam
Learning rate	10^{-3}
Weight decay	10^{-5}
Training batch size	128
Validation batch size	512
Gradient-norm clipping	100
Maximum updates	30,000
Validation frequency	Every 250 updates
Early-stopping patience	15 validation checks
Minimum relative improvement	0.001

This page summarizes the architecture, training objective, and optimization settings used for the detailed 46-channel results that follow.

Detailed Per-Channel Results for All 46 Output Channels

BOREAS supplementary material

This document reports the detailed physical-scale RMSE, MAE, and MSE for all 46 output channels, corresponding to the aggregate single-step and recursive rollout tables in the main text. The four sections use the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$.

1 Detailed Results for All 46 Output Channels

The aggregate tables in the main text report errors pooled within six physical groups. This section exposes the corresponding results for all 46 output channels, allowing within-group variation to be inspected directly. The $H = 1$ tables correspond to Table 1; the $H = 5, 10, 15$ tables correspond to Table 2. Each horizon reports physical-scale RMSE, MAE, and MSE for the same ten models and test windows.

Table 1: Detailed per-channel RMSE results corresponding to Table 1. The table reports RMSE in native physical units for all 46 output channels at $H = 1$ (5 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	Kalman- SS	Inform- er	TSMixer	TimesNet	LightTS	BOREAS	
cold_aisle_T_00	Cold-T	°C	0.4828	0.3523	0.4760	0.6808	0.4667	0.5766	0.5008	0.4205	0.4209	0.3088
cold_aisle_T_01	Cold-T	°C	0.5588	0.5134	0.5644	1.1021	1.0585	1.0574	0.5880	0.5326	0.6212	0.4851
cold_aisle_T_02	Cold-T	°C	0.4807	0.3619	0.5361	0.7635	0.5473	0.7761	0.4754	0.4365	0.4696	0.3175
cold_aisle_T_03	Cold-T	°C	0.2945	0.2504	0.3894	0.5421	0.4009	0.5170	0.3539	0.3093	0.3187	0.2208
cold_aisle_T_04	Cold-T	°C	0.2568	0.1844	0.2728	0.9071	0.9519	0.7004	0.3066	0.2286	0.2753	0.1552
cold_aisle_T_05	Cold-T	°C	0.2643	0.2255	0.3038	0.3814	0.2447	0.4826	0.2560	0.2595	0.2481	0.1954
cold_aisle_T_06	Cold-T	°C	0.6584	0.5656	0.7176	0.9930	0.8260	1.1104	0.6447	0.6979	0.7153	0.5028
cold_aisle_T_07	Cold-T	°C	0.5972	0.5256	0.7741	0.9289	0.8069	0.8092	0.5864	0.5430	0.5949	0.4918
cold_aisle_T_08	Cold-T	°C	0.3646	0.3203	0.4782	0.8671	0.7767	0.8612	0.4277	0.3399	0.3849	0.2950
cold_aisle_T_09	Cold-T	°C	0.4161	0.3714	0.4003	0.6832	0.7696	0.8171	0.4026	0.3598	0.3984	0.3481
cold_aisle_T_10	Cold-T	°C	0.3920	0.2866	0.3062	0.7734	0.3406	0.8559	0.3170	0.3100	0.3731	0.2113
cold_aisle_T_11	Cold-T	°C	0.9096	0.8814	0.8542	1.0499	0.8647	1.1515	0.8147	0.8349	0.8325	0.8741
hot_aisle_T_00	Hot-T	°C	0.2188	0.1883	0.2676	0.5507	0.6379	0.4359	0.2109	0.1850	0.1766	0.1402
hot_aisle_T_01	Hot-T	°C	0.2013	0.1383	0.2342	0.5165	0.5700	0.6762	0.1812	0.1668	0.2141	0.1134
cold_aisle_H_00	Cold-RH	%RH	1.2874	1.1011	1.4974	2.5026	1.7238	4.1560	1.5794	1.3906	1.2815	0.9191
cold_aisle_H_01	Cold-RH	%RH	1.6872	1.5653	1.7447	3.5687	3.5485	3.6130	1.7419	1.6684	1.8958	1.4762
cold_aisle_H_02	Cold-RH	%RH	1.3266	1.1027	1.2808	2.6788	1.2544	3.8019	1.5687	1.3613	1.5198	0.9711
cold_aisle_H_03	Cold-RH	%RH	0.7680	0.7756	1.3320	1.6626	1.2752	3.7771	1.5195	0.9444	1.2716	0.6381
cold_aisle_H_04	Cold-RH	%RH	0.9078	0.6117	0.8252	3.2835	3.5455	4.8935	1.4469	0.8305	1.0053	0.4498
cold_aisle_H_05	Cold-RH	%RH	0.7113	0.6380	0.8099	1.2867	0.7259	2.6799	1.5356	0.8829	1.0291	0.5483
cold_aisle_H_06	Cold-RH	%RH	1.8743	1.7192	2.1644	3.1518	2.8048	3.6041	1.9645	2.0384	2.4340	1.5099
cold_aisle_H_07	Cold-RH	%RH	1.8446	1.5944	3.0688	3.4141	3.2163	4.5093	1.7724	1.7420	2.1317	1.4414
cold_aisle_H_08	Cold-RH	%RH	0.9968	0.9322	1.6960	3.3978	2.9653	3.0288	1.2539	1.0265	1.3277	0.8364
cold_aisle_H_09	Cold-RH	%RH	1.1493	1.0467	1.1997	1.9175	2.4234	3.1143	1.3129	1.0953	1.3884	0.9631
cold_aisle_H_10	Cold-RH	%RH	0.9486	0.7974	0.8699	2.1893	0.7962	2.5679	1.2096	0.9992	1.0607	0.6199
cold_aisle_H_11	Cold-RH	%RH	2.9656	2.8957	3.0539	3.4898	3.0463	3.4414	2.9116	2.7630	2.7478	2.8609
hot_aisle_H_00	Hot-RH	%RH	0.5528	0.3197	0.5423	0.8603	1.0089	1.6453	0.4346	0.3480	0.4573	0.2467
hot_aisle_H_01	Hot-RH	%RH	0.4892	0.2863	0.6062	0.9461	0.9598	1.6757	0.5714	0.3478	0.4201	0.2352
ACU_LAT_0	ACU LAT	°C	0.6112	0.4272	0.6165	2.0322	0.5829	0.9059	0.5006	0.3351	0.4963	0.1276
ACU_LAT_1	ACU LAT	°C	0.4306	0.4357	0.6307	0.6933	0.8703	0.7638	0.3783	0.4032	0.5400	0.3616
ACU_LAT_2	ACU LAT	°C	0.3065	0.3406	0.4892	0.8696	0.7560	0.7956	0.4467	0.3844	0.5224	0.3344
ACU_LAT_3	ACU LAT	°C	0.4421	0.4126	0.5487	0.8222	0.5144	0.7939	0.3063	0.3454	0.3924	0.2393
ACU_LAT_4	ACU LAT	°C	0.4177	0.2841	0.5226	0.9514	0.5354	0.5823	0.3123	0.2666	0.3499	0.1513
ACU_LAT_5	ACU LAT	°C	0.1431	0.1378	0.3239	0.2807	0.2786	0.3355	0.2234	0.1816	0.2603	0.1325
ACU_LAT_6	ACU LAT	°C	0.2643	0.2563	0.3509	0.4769	0.6166	0.5559	0.3317	0.2634	0.4359	0.1271
ACU_LAT_7	ACU LAT	°C	0.2420	0.1852	0.4955	0.6175	0.4689	1.0883	0.2264	0.2226	0.2818	0.0950
ACU_LAT_8	ACU LAT	°C	0.5492	0.4927	0.5738	0.6834	0.6658	0.6370	0.3678	0.2317	0.4510	0.1437
ACU_EAT_0	ACU EAT	°C	0.2091	0.1657	0.3197	0.5766	0.5044	0.5409	0.1895	0.1573	0.2207	0.0947
ACU_EAT_1	ACU EAT	°C	0.1776	0.1460	0.2688	0.2185	0.3932	0.3776	0.1283	0.1436	0.1577	0.0965
ACU_EAT_2	ACU EAT	°C	0.1234	0.1193	0.2217	0.3714	0.2297	0.2843	0.2515	0.1288	0.1457	0.0988
ACU_EAT_3	ACU EAT	°C	0.1585	0.1385	0.3159	0.5030	0.3433	0.5613	0.2108	0.1608	0.1755	0.0977
ACU_EAT_4	ACU EAT	°C	0.1203	0.1102	0.1490	0.1729	0.1726	0.1693	0.1056	0.0937	0.0962	0.0733
ACU_EAT_5	ACU EAT	°C	0.1868	0.1477	0.3494	0.3743	0.3424	0.3502	0.1715	0.1490	0.1770	0.1025
ACU_EAT_6	ACU EAT	°C	0.1913	0.1578	0.1947	0.2680	0.1831	0.2091	0.1057	0.1118	0.1535	0.0792
ACU_EAT_7	ACU EAT	°C	0.2427	0.1892	0.2390	0.3877	0.2964	0.2352	0.1237	0.1002	0.1384	0.0814
ACU_EAT_8	ACU EAT	°C	0.3269	0.2557	0.4466	0.5881	0.3912	0.3810	0.1806	0.1014	0.1858	0.0779

Table 2: Detailed per-channel MAE results corresponding to Table 1. The table reports MAE in native physical units for all 46 output channels at $H = 1$ (5 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS	Ridge	Subspace-	Kalman-	Informer	TSMixer	TimesNet	LightTS	BOREAS	
			ARX	ARX		SS						SS
cold_aisle_T_00	Cold-T	°C	0.3896	0.2351	0.3614	0.4892	0.3007	0.3896	0.4063	0.2855	0.3021	0.1579
cold_aisle_T_01	Cold-T	°C	0.3749	0.2768	0.3519	0.7169	0.7277	0.8004	0.3941	0.2988	0.3874	0.2349
cold_aisle_T_02	Cold-T	°C	0.3611	0.2462	0.3922	0.5056	0.4099	0.5831	0.3929	0.2792	0.3424	0.1691
cold_aisle_T_03	Cold-T	°C	0.2204	0.1591	0.2899	0.3636	0.3100	0.3454	0.2636	0.1871	0.1978	0.1189
cold_aisle_T_04	Cold-T	°C	0.1861	0.1330	0.1975	0.5759	0.6354	0.4593	0.2037	0.1444	0.1812	0.1031
cold_aisle_T_05	Cold-T	°C	0.1914	0.1598	0.2294	0.2685	0.1797	0.3688	0.1797	0.1770	0.1633	0.1212
cold_aisle_T_06	Cold-T	°C	0.4611	0.3533	0.5696	0.7897	0.6497	0.8558	0.4899	0.4344	0.4383	0.2708
cold_aisle_T_07	Cold-T	°C	0.3947	0.3271	0.5815	0.6994	0.5942	0.6100	0.4265	0.3458	0.4023	0.2737
cold_aisle_T_08	Cold-T	°C	0.2392	0.1926	0.3893	0.6509	0.6165	0.6409	0.3372	0.2130	0.2677	0.1627
cold_aisle_T_09	Cold-T	°C	0.2620	0.2111	0.2663	0.5361	0.6069	0.5914	0.2735	0.2111	0.2411	0.1758
cold_aisle_T_10	Cold-T	°C	0.2758	0.1913	0.2056	0.5755	0.2406	0.6394	0.2247	0.2234	0.2596	0.1434
cold_aisle_T_11	Cold-T	°C	0.4460	0.3920	0.3935	0.6603	0.3950	0.7619	0.3608	0.3773	0.4330	0.3205
hot_aisle_T_00	Hot-T	°C	0.1707	0.1446	0.2142	0.3973	0.4557	0.2894	0.1582	0.1189	0.1200	0.0842
hot_aisle_T_01	Hot-T	°C	0.1483	0.1102	0.1789	0.4228	0.4648	0.5116	0.1269	0.1272	0.1536	0.0778
cold_aisle_H_00	Cold-RH	%RH	0.9868	0.7702	1.0523	1.8372	1.2856	2.5972	1.1022	1.0450	0.8686	0.4478
cold_aisle_H_01	Cold-RH	%RH	1.0138	0.8123	1.0499	2.4191	2.2964	2.4735	1.1747	0.9342	1.2397	0.7383
cold_aisle_H_02	Cold-RH	%RH	0.9673	0.7287	0.9637	1.9146	0.9228	2.4820	1.2982	0.8996	1.0378	0.5127
cold_aisle_H_03	Cold-RH	%RH	0.5147	0.5027	0.9355	1.0714	0.9781	2.2241	1.0559	0.5855	0.8925	0.3436
cold_aisle_H_04	Cold-RH	%RH	0.6831	0.4497	0.5875	2.2331	2.5405	3.2944	1.1086	0.5858	0.6786	0.3081
cold_aisle_H_05	Cold-RH	%RH	0.5451	0.4711	0.6410	0.9532	0.5577	1.6903	1.1173	0.6464	0.7582	0.3551
cold_aisle_H_06	Cold-RH	%RH	1.3236	1.0729	1.7470	2.4915	2.1538	2.7237	1.4888	1.3271	1.7694	0.7945
cold_aisle_H_07	Cold-RH	%RH	1.2354	1.0269	2.3378	2.7006	2.3810	3.0183	1.2991	1.1422	1.5299	0.8140
cold_aisle_H_08	Cold-RH	%RH	0.6447	0.5799	1.3447	2.4442	2.3692	2.1216	0.9422	0.7277	0.9475	0.4679
cold_aisle_H_09	Cold-RH	%RH	0.7281	0.6199	0.7528	1.5128	1.9080	1.9824	0.8745	0.7405	0.8436	0.5015
cold_aisle_H_10	Cold-RH	%RH	0.6537	0.5356	0.6207	1.6511	0.5393	1.6725	0.9116	0.7295	0.7950	0.4249
cold_aisle_H_11	Cold-RH	%RH	1.3572	1.2913	1.2718	2.0237	1.3301	1.9576	1.5424	1.2844	1.6117	1.0419
hot_aisle_H_00	Hot-RH	%RH	0.4393	0.2444	0.4485	0.6973	0.7799	0.9836	0.3226	0.2339	0.3469	0.1537
hot_aisle_H_01	Hot-RH	%RH	0.3563	0.2214	0.4542	0.6792	0.6904	1.1649	0.4256	0.2617	0.3181	0.1702
ACU_LAT_0	ACU LAT	°C	0.5089	0.3423	0.4852	1.5905	0.4470	0.7067	0.3580	0.2390	0.3571	0.0630
ACU_LAT_1	ACU LAT	°C	0.2554	0.1822	0.3485	0.4555	0.6377	0.5410	0.2141	0.2203	0.2081	0.1138
ACU_LAT_2	ACU LAT	°C	0.1375	0.1362	0.3018	0.6083	0.5280	0.5725	0.2887	0.1686	0.2655	0.1162
ACU_LAT_3	ACU LAT	°C	0.3225	0.2753	0.3569	0.6566	0.3718	0.5562	0.1938	0.2373	0.2366	0.1094
ACU_LAT_4	ACU LAT	°C	0.3416	0.2256	0.4058	0.7030	0.4158	0.4735	0.2413	0.1894	0.2814	0.0933
ACU_LAT_5	ACU LAT	°C	0.1009	0.0823	0.2609	0.1897	0.1933	0.2278	0.1575	0.1085	0.1707	0.0685
ACU_LAT_6	ACU LAT	°C	0.2114	0.2022	0.2571	0.3566	0.4950	0.3979	0.2645	0.1862	0.3240	0.0831
ACU_LAT_7	ACU LAT	°C	0.1552	0.1147	0.3608	0.4727	0.3271	0.7651	0.1556	0.1561	0.1781	0.0431
ACU_LAT_8	ACU LAT	°C	0.3790	0.3191	0.4370	0.5523	0.4422	0.4867	0.2696	0.1581	0.3185	0.0807
ACU_EAT_0	ACU EAT	°C	0.1661	0.1284	0.2729	0.4918	0.4132	0.4147	0.1393	0.1136	0.1597	0.0615
ACU_EAT_1	ACU EAT	°C	0.1303	0.1051	0.1886	0.1532	0.2816	0.2474	0.0930	0.0978	0.1044	0.0617
ACU_EAT_2	ACU EAT	°C	0.0887	0.0745	0.1827	0.2856	0.1663	0.1940	0.1814	0.0781	0.1020	0.0553
ACU_EAT_3	ACU EAT	°C	0.1197	0.0935	0.2223	0.3691	0.2517	0.3629	0.1458	0.1057	0.1260	0.0555
ACU_EAT_4	ACU EAT	°C	0.1000	0.0889	0.1180	0.1354	0.1437	0.1389	0.0834	0.0716	0.0743	0.0532
ACU_EAT_5	ACU EAT	°C	0.1493	0.1127	0.2204	0.2318	0.2323	0.2496	0.1146	0.1094	0.1306	0.0685
ACU_EAT_6	ACU EAT	°C	0.1377	0.1138	0.1322	0.1879	0.1292	0.1397	0.0762	0.0823	0.1078	0.0537
ACU_EAT_7	ACU EAT	°C	0.1743	0.1298	0.1686	0.2782	0.2031	0.1555	0.0900	0.0702	0.1011	0.0552
ACU_EAT_8	ACU EAT	°C	0.2187	0.1687	0.2971	0.3949	0.2585	0.2584	0.1304	0.0719	0.1323	0.0531

Table 3: Detailed per-channel MSE results corresponding to Table 1. The table reports MSE in squared native units for all 46 output channels at $H = 1$ (5 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- Ridge-		Subspace- Kalman-	Informer	TSMixer	TimesNet	LightTS	BOREAS		
			ARX	ARX								
cold_aisle_T_00	Cold-T	°C	0.2331	0.1241	0.2265	0.4636	0.2178	0.3325	0.2508	0.1768	0.1772	0.0953
cold_aisle_T_01	Cold-T	°C	0.3122	0.2635	0.3185	1.2147	1.1205	1.1182	0.3458	0.2836	0.3859	0.2353
cold_aisle_T_02	Cold-T	°C	0.2311	0.1310	0.2874	0.5829	0.2996	0.6023	0.2260	0.1905	0.2205	0.1008
cold_aisle_T_03	Cold-T	°C	0.0867	0.0627	0.1516	0.2939	0.1607	0.2673	0.1252	0.0957	0.1016	0.0488
cold_aisle_T_04	Cold-T	°C	0.0659	0.0340	0.0744	0.8228	0.9061	0.4905	0.0940	0.0523	0.0758	0.0241
cold_aisle_T_05	Cold-T	°C	0.0699	0.0508	0.0923	0.1455	0.0599	0.2329	0.0655	0.0674	0.0616	0.0382
cold_aisle_T_06	Cold-T	°C	0.4335	0.3199	0.5150	0.9861	0.6823	1.2331	0.4156	0.4870	0.5117	0.2528
cold_aisle_T_07	Cold-T	°C	0.3566	0.2763	0.5992	0.8629	0.6511	0.6549	0.3439	0.2949	0.3539	0.2419
cold_aisle_T_08	Cold-T	°C	0.1330	0.1026	0.2287	0.7518	0.6033	0.7417	0.1829	0.1155	0.1481	0.0870
cold_aisle_T_09	Cold-T	°C	0.1731	0.1379	0.1602	0.4668	0.5923	0.6676	0.1621	0.1294	0.1587	0.1212
cold_aisle_T_10	Cold-T	°C	0.1537	0.0821	0.0938	0.5981	0.1160	0.7325	0.1005	0.0961	0.1392	0.0446
cold_aisle_T_11	Cold-T	°C	0.8274	0.7768	0.7297	1.1024	0.7477	1.3260	0.6638	0.6971	0.6931	0.7641
hot_aisle_T_00	Hot-T	°C	0.0479	0.0355	0.0716	0.3033	0.4069	0.1901	0.0445	0.0342	0.0312	0.0197
hot_aisle_T_01	Hot-T	°C	0.0405	0.0191	0.0548	0.2668	0.3249	0.4572	0.0328	0.0278	0.0459	0.0129
cold_aisle_H_00	Cold-RH	%RH	1.6574	1.2125	2.2423	6.2628	2.9716	17.2726	2.4945	1.9338	1.6422	0.8448
cold_aisle_H_01	Cold-RH	%RH	2.8466	2.4501	3.0441	12.7356	12.5919	13.0535	3.0342	2.7834	3.5940	2.1791
cold_aisle_H_02	Cold-RH	%RH	1.7599	1.2159	1.6405	7.1757	1.5735	14.4547	2.4607	1.8532	2.3099	0.9430
cold_aisle_H_03	Cold-RH	%RH	0.5898	0.6016	1.7741	2.7641	1.6260	14.2667	2.3088	0.8918	1.6169	0.4071
cold_aisle_H_04	Cold-RH	%RH	0.8242	0.3742	0.6810	10.7812	12.5703	23.9464	2.0936	0.6897	1.0107	0.2023
cold_aisle_H_05	Cold-RH	%RH	0.5059	0.4071	0.6560	1.6556	0.5269	7.1820	2.3579	0.7795	1.0591	0.3007
cold_aisle_H_06	Cold-RH	%RH	3.5130	2.9555	4.6847	9.9339	7.8667	12.9893	3.8593	4.1550	5.9241	2.2799
cold_aisle_H_07	Cold-RH	%RH	3.4027	2.5422	9.4175	11.6559	10.3446	20.3336	3.1416	3.0347	4.5442	2.0775
cold_aisle_H_08	Cold-RH	%RH	0.9936	0.8690	2.8765	11.5451	8.7929	9.1738	1.5724	1.0537	1.7627	0.6995
cold_aisle_H_09	Cold-RH	%RH	1.3209	1.0956	1.4392	3.6767	5.8727	9.6991	1.7237	1.1997	1.9275	0.9276
cold_aisle_H_10	Cold-RH	%RH	0.8999	0.6358	0.7568	4.7931	0.6339	6.5940	1.4630	0.9983	1.1251	0.3843
cold_aisle_H_11	Cold-RH	%RH	8.7950	8.3848	9.3261	12.1788	9.2802	11.8430	8.4774	7.6344	7.5504	8.1850
hot_aisle_H_00	Hot-RH	%RH	0.3056	0.1022	0.2941	0.7402	1.0180	2.7070	0.1889	0.1211	0.2092	0.0609
hot_aisle_H_01	Hot-RH	%RH	0.2394	0.0820	0.3674	0.8952	0.9212	2.8078	0.3265	0.1209	0.1765	0.0553
ACU_LAT_0	ACU LAT	°C	0.3736	0.1825	0.3801	4.1297	0.3398	0.8207	0.2506	0.1123	0.2463	0.0163
ACU_LAT_1	ACU LAT	°C	0.1854	0.1899	0.3978	0.4806	0.7575	0.5835	0.1431	0.1626	0.2916	0.1308
ACU_LAT_2	ACU LAT	°C	0.0939	0.1160	0.2393	0.7562	0.5715	0.6330	0.1996	0.1478	0.2729	0.1118
ACU_LAT_3	ACU LAT	°C	0.1955	0.1702	0.3010	0.6760	0.2646	0.6304	0.0938	0.1193	0.1539	0.0573
ACU_LAT_4	ACU LAT	°C	0.1745	0.0807	0.2731	0.9051	0.2867	0.3390	0.0975	0.0711	0.1225	0.0229
ACU_LAT_5	ACU LAT	°C	0.0205	0.0190	0.1049	0.0788	0.0776	0.1126	0.0499	0.0330	0.0678	0.0176
ACU_LAT_6	ACU LAT	°C	0.0698	0.0657	0.1231	0.2275	0.3802	0.3090	0.1100	0.0694	0.1900	0.0161
ACU_LAT_7	ACU LAT	°C	0.0586	0.0343	0.2455	0.3813	0.2199	1.1844	0.0513	0.0496	0.0794	0.0090
ACU_LAT_8	ACU LAT	°C	0.3016	0.2427	0.3293	0.4671	0.4432	0.4058	0.1353	0.0537	0.2034	0.0206
ACU_EAT_0	ACU EAT	°C	0.0437	0.0275	0.1022	0.3325	0.2544	0.2925	0.0359	0.0247	0.0487	0.0090
ACU_EAT_1	ACU EAT	°C	0.0315	0.0213	0.0723	0.0477	0.1546	0.1426	0.0165	0.0206	0.0249	0.0093
ACU_EAT_2	ACU EAT	°C	0.0152	0.0142	0.0492	0.1379	0.0528	0.0809	0.0633	0.0166	0.0212	0.0098
ACU_EAT_3	ACU EAT	°C	0.0251	0.0192	0.0998	0.2530	0.1178	0.3150	0.0444	0.0258	0.0308	0.0095
ACU_EAT_4	ACU EAT	°C	0.0145	0.0121	0.0222	0.0299	0.0298	0.0287	0.0111	0.0088	0.0093	0.0054
ACU_EAT_5	ACU EAT	°C	0.0349	0.0218	0.1221	0.1401	0.1173	0.1226	0.0294	0.0222	0.0313	0.0105
ACU_EAT_6	ACU EAT	°C	0.0366	0.0249	0.0379	0.0718	0.0335	0.0437	0.0112	0.0125	0.0236	0.0063
ACU_EAT_7	ACU EAT	°C	0.0589	0.0358	0.0571	0.1503	0.0878	0.0553	0.0153	0.0100	0.0192	0.0066
ACU_EAT_8	ACU EAT	°C	0.1068	0.0654	0.1995	0.3459	0.1530	0.1452	0.0326	0.0103	0.0345	0.0061

Table 4: Detailed per-channel RMSE results corresponding to Table 2. The table reports RMSE in native physical units for all 46 output channels at $H = 5$ (25 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	Kalman- SS	Inform- er	TSMixer	TimesNet	LightTS	BOREAS	
cold_aisle_T_00	Cold-T	°C	0.9505	0.7203	0.8609	0.8126	0.7948	0.7473	1.0913	0.8514	0.9513	0.6854
cold_aisle_T_01	Cold-T	°C	1.1607	1.0976	1.1254	1.2778	1.3148	1.4412	1.2028	1.0941	1.1049	1.0906
cold_aisle_T_02	Cold-T	°C	0.9885	0.8144	1.0027	0.9302	0.9968	1.1649	0.9702	0.7679	0.8530	0.6986
cold_aisle_T_03	Cold-T	°C	0.5961	0.5256	0.6537	0.6232	0.6385	0.6639	0.7070	0.5484	0.5643	0.4779
cold_aisle_T_04	Cold-T	°C	0.4736	0.3888	0.4899	0.9504	0.9595	0.9709	0.6053	0.3496	0.4265	0.2621
cold_aisle_T_05	Cold-T	°C	0.4970	0.4501	0.5979	0.4516	0.4866	0.6460	0.4174	0.3990	0.3770	0.3317
cold_aisle_T_06	Cold-T	°C	1.2703	1.1919	1.1418	1.1329	1.1355	1.4730	1.2538	1.1872	1.0682	1.1152
cold_aisle_T_07	Cold-T	°C	1.2241	1.1070	1.0434	0.9577	1.0623	1.0219	1.1895	1.0553	0.9648	1.0383
cold_aisle_T_08	Cold-T	°C	0.6947	0.6306	0.7697	0.8917	0.8738	1.1023	0.8343	0.5242	0.5313	0.5760
cold_aisle_T_09	Cold-T	°C	0.8481	0.7357	0.7030	0.7211	0.8832	1.0781	0.7605	0.6255	0.5515	0.6631
cold_aisle_T_10	Cold-T	°C	0.8548	0.6584	0.6680	0.8501	0.5553	1.1284	0.5375	0.4876	0.6302	0.3851
cold_aisle_T_11	Cold-T	°C	1.5777	1.5181	1.4629	1.1088	1.4552	1.5879	1.2561	1.3045	1.1307	1.4998
hot_aisle_T_00	Hot-T	°C	0.4784	0.4246	0.4893	0.5911	0.6972	0.5809	0.3610	0.2808	0.2795	0.2488
hot_aisle_T_01	Hot-T	°C	0.4809	0.3653	0.4564	0.5585	0.6087	0.8865	0.3534	0.3344	0.3617	0.2560
cold_aisle_H_00	Cold-RH	%RH	2.7176	2.4074	2.9873	2.9367	2.7530	6.1230	3.4580	2.9832	3.0345	2.0684
cold_aisle_H_01	Cold-RH	%RH	3.6302	3.4773	3.6481	4.1259	4.3933	5.6638	3.5258	3.5865	3.5947	3.4056
cold_aisle_H_02	Cold-RH	%RH	2.9448	2.6180	2.7186	3.0990	2.7739	5.7666	3.4164	2.4395	2.6868	2.1526
cold_aisle_H_03	Cold-RH	%RH	1.6438	1.7611	2.2430	1.9356	2.1096	6.0521	3.0076	1.8537	1.9895	1.4018
cold_aisle_H_04	Cold-RH	%RH	2.0212	1.4726	1.5830	3.4384	3.5457	7.1325	3.0890	1.5771	1.7349	0.7677
cold_aisle_H_05	Cold-RH	%RH	1.4827	1.3402	1.7623	1.5122	1.4706	3.8702	2.8704	1.3847	1.6053	0.9422
cold_aisle_H_06	Cold-RH	%RH	3.8835	3.7349	3.7040	3.5971	3.8070	5.3525	3.6649	3.9759	3.8426	3.4081
cold_aisle_H_07	Cold-RH	%RH	3.8815	3.5251	3.8112	3.4472	3.8986	5.9264	3.2915	3.6532	3.5612	3.1063
cold_aisle_H_08	Cold-RH	%RH	1.9849	1.8742	2.8016	3.4731	3.2350	4.4538	2.4390	1.9583	2.0461	1.6461
cold_aisle_H_09	Cold-RH	%RH	2.3491	2.1108	2.2384	2.0917	2.8775	4.4015	2.4281	2.3109	2.0686	1.8238
cold_aisle_H_10	Cold-RH	%RH	1.9987	1.7777	1.9293	2.4597	1.5442	3.6549	2.4196	1.6993	1.8717	1.1365
cold_aisle_H_11	Cold-RH	%RH	5.3466	5.1580	5.4047	3.7386	5.2960	5.1797	4.8604	4.4542	3.9344	5.0605
hot_aisle_H_00	Hot-RH	%RH	1.1465	0.7548	0.8277	0.8996	1.0889	2.3817	0.8968	0.6650	0.7230	0.4610
hot_aisle_H_01	Hot-RH	%RH	1.0608	0.8005	0.9332	1.0671	1.0743	2.5410	1.1971	0.7530	0.7954	0.5629
ACU_LAT_0	ACU LAT	°C	1.1933	0.9001	1.1602	2.1470	1.0821	1.4301	1.0449	0.6407	0.8950	0.2401
ACU_LAT_1	ACU LAT	°C	0.8099	0.7432	0.9019	0.7266	0.9650	0.8446	0.5767	0.5935	0.6860	0.5624
ACU_LAT_2	ACU LAT	°C	0.5593	0.6353	0.7630	0.9045	0.8584	0.9648	0.7125	0.5822	0.7051	0.6012
ACU_LAT_3	ACU LAT	°C	0.9951	0.8789	0.9707	0.8986	0.8633	1.1235	0.5772	0.7156	0.6198	0.3921
ACU_LAT_4	ACU LAT	°C	0.8656	0.6717	0.9422	1.0355	0.8370	0.7941	0.5808	0.4747	0.7561	0.2411
ACU_LAT_5	ACU LAT	°C	0.3021	0.2759	0.4497	0.3185	0.3842	0.4122	0.3898	0.2884	0.3756	0.2318
ACU_LAT_6	ACU LAT	°C	0.6173	0.6185	0.5268	0.5883	0.5962	0.6807	0.6125	0.4745	0.7383	0.1986
ACU_LAT_7	ACU LAT	°C	0.6031	0.4714	1.0498	0.6425	0.7759	1.1999	0.4997	0.3828	0.4968	0.1547
ACU_LAT_8	ACU LAT	°C	1.2799	1.1921	1.2459	0.8804	1.1185	0.9551	0.7258	0.3551	0.8378	0.2311
ACU_EAT_0	ACU EAT	°C	0.4855	0.3958	0.6348	0.6307	0.5714	0.5840	0.2905	0.2931	0.3538	0.1675
ACU_EAT_1	ACU EAT	°C	0.4408	0.3579	0.4821	0.2471	0.5215	0.4325	0.2327	0.2674	0.2556	0.1758
ACU_EAT_2	ACU EAT	°C	0.3341	0.2949	0.4267	0.4142	0.3915	0.4464	0.4860	0.2712	0.2420	0.1936
ACU_EAT_3	ACU EAT	°C	0.4346	0.3674	0.5512	0.5592	0.5007	0.6824	0.4648	0.3551	0.2818	0.1925
ACU_EAT_4	ACU EAT	°C	0.3299	0.2865	0.3230	0.2112	0.2895	0.2460	0.1970	0.1715	0.1736	0.1121
ACU_EAT_5	ACU EAT	°C	0.4785	0.3679	0.4919	0.4074	0.4641	0.4449	0.2647	0.2615	0.2921	0.1546
ACU_EAT_6	ACU EAT	°C	0.5024	0.4207	0.4567	0.3241	0.4030	0.3177	0.2242	0.2158	0.2892	0.1299
ACU_EAT_7	ACU EAT	°C	0.6307	0.5103	0.5781	0.4477	0.5546	0.3462	0.2624	0.1878	0.2752	0.1375
ACU_EAT_8	ACU EAT	°C	0.8124	0.6900	0.8267	0.6833	0.7002	0.4928	0.3627	0.1871	0.3905	0.1418

Table 5: Detailed per-channel MAE results corresponding to Table 2. The table reports MAE in native physical units for all 46 output channels at $H = 5$ (25 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	Kalman- SS	Inform- er	TSMixer	TimesNet	LightTS	BOREAS
cold_aisle_T_00	Cold-T	°C	0.7437	0.4985	0.6141	0.5519	0.5668	0.4777	0.8590	0.5908	0.3762
cold_aisle_T_01	Cold-T	°C	0.8063	0.6485	0.6668	0.8363	0.8469	1.0735	0.8359	0.6546	0.5648
cold_aisle_T_02	Cold-T	°C	0.7073	0.5906	0.7363	0.6444	0.7575	0.8712	0.7907	0.4925	0.3645
cold_aisle_T_03	Cold-T	°C	0.4432	0.3494	0.4632	0.4046	0.4589	0.4773	0.5149	0.3403	0.3545
cold_aisle_T_04	Cold-T	°C	0.3379	0.2678	0.3453	0.5956	0.6353	0.6417	0.3662	0.2176	0.2699
cold_aisle_T_05	Cold-T	°C	0.3406	0.3216	0.4430	0.3110	0.3540	0.5027	0.3026	0.2714	0.2551
cold_aisle_T_06	Cold-T	°C	0.8872	0.7912	0.8279	0.9038	0.8436	1.1103	0.9228	0.8047	0.6753
cold_aisle_T_07	Cold-T	°C	0.8117	0.7176	0.7447	0.7124	0.7595	0.7861	0.8203	0.7079	0.6574
cold_aisle_T_08	Cold-T	°C	0.4518	0.3851	0.5705	0.6673	0.6813	0.8298	0.6166	0.3506	0.3727
cold_aisle_T_09	Cold-T	°C	0.5357	0.4368	0.4531	0.5609	0.6869	0.7994	0.4953	0.3922	0.3444
cold_aisle_T_10	Cold-T	°C	0.5854	0.4158	0.3955	0.6256	0.3715	0.8633	0.3713	0.3468	0.4453
cold_aisle_T_11	Cold-T	°C	0.9218	0.8192	0.7537	0.7112	0.7481	1.0773	0.6245	0.6372	0.6171
hot_aisle_T_00	Hot-T	°C	0.3523	0.3140	0.3712	0.4295	0.5116	0.3928	0.2632	0.1753	0.1444
hot_aisle_T_01	Hot-T	°C	0.3303	0.2648	0.3267	0.4410	0.4548	0.6813	0.2242	0.2317	0.2393
cold_aisle_H_00	Cold-RH	%RH	1.9939	1.7482	2.1578	2.0518	2.0279	3.7287	2.4990	2.1783	2.0673
cold_aisle_H_01	Cold-RH	%RH	2.2969	1.9292	2.1717	2.7515	2.7890	3.8977	2.1984	2.2398	2.2862
cold_aisle_H_02	Cold-RH	%RH	2.1036	1.8759	1.9742	2.1814	2.0718	3.8351	2.7589	1.5440	1.8786
cold_aisle_H_03	Cold-RH	%RH	1.1255	1.1768	1.5704	1.2382	1.4987	3.6178	2.0814	1.1771	1.4361
cold_aisle_H_04	Cold-RH	%RH	1.4269	1.0091	1.1303	2.2688	2.4911	4.8092	2.2153	1.0510	1.1775
cold_aisle_H_05	Cold-RH	%RH	1.1123	0.9894	1.3329	1.0974	1.0883	2.4680	2.0001	0.9987	1.2137
cold_aisle_H_06	Cold-RH	%RH	2.8039	2.4767	2.7068	2.8689	2.7896	3.8631	2.7452	2.7373	2.9046
cold_aisle_H_07	Cold-RH	%RH	2.6097	2.3506	2.7924	2.7224	2.8712	3.9548	2.2858	2.4523	2.5897
cold_aisle_H_08	Cold-RH	%RH	1.2913	1.1766	2.0947	2.5297	2.5739	2.9983	1.7724	1.3739	1.4417
cold_aisle_H_09	Cold-RH	%RH	1.5200	1.2982	1.4574	1.6348	2.2349	2.8380	1.5425	1.5679	1.3153
cold_aisle_H_10	Cold-RH	%RH	1.2742	1.1044	1.1806	1.8214	1.0404	2.4155	1.6994	1.2310	1.3979
cold_aisle_H_11	Cold-RH	%RH	2.8897	2.7537	2.7559	2.2722	2.7876	3.1859	2.8662	2.2827	2.4617
hot_aisle_H_00	Hot-RH	%RH	0.8477	0.5468	0.6547	0.7230	0.8395	1.3985	0.6419	0.4233	0.5553
hot_aisle_H_01	Hot-RH	%RH	0.7298	0.5590	0.6983	0.7371	0.7926	1.7457	0.8561	0.5179	0.5732
ACU_LAT_0	ACU LAT	°C	0.9797	0.7337	0.9159	1.6784	0.8336	1.1377	0.7353	0.4448	0.6275
ACU_LAT_1	ACU LAT	°C	0.5567	0.3830	0.5494	0.4797	0.6833	0.6153	0.3479	0.3327	0.2858
ACU_LAT_2	ACU LAT	°C	0.2737	0.2760	0.4837	0.6244	0.5831	0.6656	0.4686	0.2700	0.3646
ACU_LAT_3	ACU LAT	°C	0.7011	0.6218	0.6014	0.6875	0.5150	0.7981	0.3891	0.4917	0.4074
ACU_LAT_4	ACU LAT	°C	0.7095	0.5268	0.7238	0.7800	0.6377	0.6321	0.4105	0.3226	0.5906
ACU_LAT_5	ACU LAT	°C	0.2126	0.1696	0.3440	0.2144	0.2831	0.2903	0.2708	0.1875	0.2559
ACU_LAT_6	ACU LAT	°C	0.4660	0.4596	0.3957	0.4466	0.4689	0.5049	0.4509	0.3146	0.5294
ACU_LAT_7	ACU LAT	°C	0.3480	0.2770	0.7148	0.4792	0.5491	0.8986	0.3527	0.2642	0.3314
ACU_LAT_8	ACU LAT	°C	0.8020	0.6902	0.8402	0.7046	0.7867	0.6879	0.5202	0.2539	0.5901
ACU_EAT_0	ACU EAT	°C	0.3730	0.2968	0.5157	0.5372	0.4866	0.4667	0.2079	0.2004	0.2643
ACU_EAT_1	ACU EAT	°C	0.3021	0.2453	0.3370	0.1701	0.3766	0.3021	0.1488	0.1770	0.1630
ACU_EAT_2	ACU EAT	°C	0.2247	0.1701	0.3057	0.3245	0.2539	0.3053	0.3366	0.1537	0.1567
ACU_EAT_3	ACU EAT	°C	0.3033	0.2403	0.3983	0.4195	0.3544	0.4481	0.2999	0.2259	0.1909
ACU_EAT_4	ACU EAT	°C	0.2561	0.2161	0.2488	0.1622	0.2385	0.2023	0.1497	0.1284	0.1377
ACU_EAT_5	ACU EAT	°C	0.3594	0.2656	0.3628	0.2628	0.3572	0.3030	0.1816	0.1891	0.2107
ACU_EAT_6	ACU EAT	°C	0.3277	0.2734	0.2864	0.2205	0.2740	0.2181	0.1546	0.1510	0.2014
ACU_EAT_7	ACU EAT	°C	0.4022	0.3066	0.3527	0.3073	0.3588	0.2397	0.1805	0.1289	0.1940
ACU_EAT_8	ACU EAT	°C	0.4898	0.4017	0.4971	0.4437	0.4041	0.3393	0.2450	0.1259	0.2598

Table 6: Detailed per-channel MSE results corresponding to Table 2. The table reports MSE in squared native units for all 46 output channels at $H = 5$ (25 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	SS	Kalman- SS	Informer	TSMixer	TimesNet	LightTS	BOREAS
cold_aisle_T_00	Cold-T	°C	0.9035	0.5189	0.7412	0.6603	0.6317	0.5584	1.1910	0.7249	0.9049	0.4698
cold_aisle_T_01	Cold-T	°C	1.3472	1.2047	1.2666	1.6329	1.7287	2.0770	1.4467	1.1970	1.2208	1.1894
cold_aisle_T_02	Cold-T	°C	0.9772	0.6632	1.0054	0.8652	0.9936	1.3571	0.9412	0.5896	0.7277	0.4880
cold_aisle_T_03	Cold-T	°C	0.3553	0.2762	0.4273	0.3884	0.4076	0.4407	0.4999	0.3007	0.3184	0.2284
cold_aisle_T_04	Cold-T	°C	0.2243	0.1511	0.2400	0.9033	0.9207	0.9426	0.3664	0.1223	0.1819	0.0687
cold_aisle_T_05	Cold-T	°C	0.2470	0.2026	0.3575	0.2039	0.2368	0.4173	0.1742	0.1592	0.1421	0.1100
cold_aisle_T_06	Cold-T	°C	1.6137	1.4207	1.3037	1.2834	1.2894	2.1697	1.5721	1.4095	1.1410	1.2436
cold_aisle_T_07	Cold-T	°C	1.4984	1.2254	1.0887	0.9171	1.1284	1.0443	1.4148	1.1136	0.9309	1.0781
cold_aisle_T_08	Cold-T	°C	0.4826	0.3976	0.5924	0.7951	0.7635	1.2151	0.6961	0.2748	0.2823	0.3317
cold_aisle_T_09	Cold-T	°C	0.7193	0.5413	0.4942	0.5200	0.7801	1.1623	0.5784	0.3913	0.3042	0.4397
cold_aisle_T_10	Cold-T	°C	0.7308	0.4335	0.4462	0.7227	0.3084	1.2732	0.2889	0.2378	0.3971	0.1483
cold_aisle_T_11	Cold-T	°C	2.4890	2.3045	2.1401	1.2294	2.1175	2.5215	1.5779	1.7017	1.2785	2.2494
hot_aisle_T_00	Hot-T	°C	0.2288	0.1803	0.2394	0.3494	0.4861	0.3374	0.1303	0.0788	0.0781	0.0619
hot_aisle_T_01	Hot-T	°C	0.2313	0.1335	0.2083	0.3119	0.3705	0.7858	0.1249	0.1118	0.1308	0.0655
cold_aisle_H_00	Cold-RH	%RH	7.3851	5.7957	8.9237	8.6242	7.5791	37.4914	11.9578	8.8995	9.2080	4.2783
cold_aisle_H_01	Cold-RH	%RH	13.1784	12.0919	13.3084	17.0232	19.3015	32.0791	12.4311	12.8632	12.9222	11.5978
cold_aisle_H_02	Cold-RH	%RH	8.6721	6.8539	7.3907	9.6041	7.6947	33.2535	11.6721	5.9511	7.2188	4.6336
cold_aisle_H_03	Cold-RH	%RH	2.7021	3.1016	5.0310	3.7464	4.4504	36.6276	9.0459	3.4361	3.9582	1.9652
cold_aisle_H_04	Cold-RH	%RH	4.0851	2.1686	2.5060	11.8223	12.5719	50.8721	9.5422	2.4872	3.0098	0.5894
cold_aisle_H_05	Cold-RH	%RH	2.1985	1.7962	3.1057	2.2868	2.1628	14.9781	8.2393	1.9173	2.5769	0.8878
cold_aisle_H_06	Cold-RH	%RH	15.0819	13.9495	13.7198	12.9391	14.4931	28.6496	13.4318	15.8080	14.7656	11.6152
cold_aisle_H_07	Cold-RH	%RH	15.0663	12.4262	14.5251	11.8834	15.1989	35.1219	10.8338	13.3457	12.6823	9.6489
cold_aisle_H_08	Cold-RH	%RH	3.9399	3.5128	7.8492	12.0625	10.4653	19.8365	5.9486	3.8349	4.1865	2.7095
cold_aisle_H_09	Cold-RH	%RH	5.5183	4.4556	5.0104	4.3751	8.2802	19.3736	5.8956	5.3403	4.2792	3.3262
cold_aisle_H_10	Cold-RH	%RH	3.9950	3.1602	3.7223	6.0504	2.3844	13.3583	5.8546	2.8875	3.5033	1.2916
cold_aisle_H_11	Cold-RH	%RH	28.5858	26.6047	29.2112	13.9768	28.0472	26.8298	23.6232	19.8398	15.4799	25.6091
hot_aisle_H_00	Hot-RH	%RH	1.3144	0.5697	0.6851	0.8092	1.1857	5.6726	0.8042	0.4422	0.5228	0.2125
hot_aisle_H_01	Hot-RH	%RH	1.1252	0.6408	0.8709	1.1386	1.1542	6.4567	1.4331	0.5670	0.6327	0.3169
ACU_LAT_0	ACU LAT	°C	1.4239	0.8101	1.3460	4.6094	1.1709	2.0453	1.0919	0.4105	0.8010	0.0576
ACU_LAT_1	ACU LAT	°C	0.6559	0.5523	0.8134	0.5279	0.9312	0.7134	0.3326	0.3522	0.4707	0.3162
ACU_LAT_2	ACU LAT	°C	0.3128	0.4036	0.5821	0.8181	0.7368	0.9308	0.5077	0.3389	0.4971	0.3614
ACU_LAT_3	ACU LAT	°C	0.9903	0.7725	0.9423	0.8076	0.7453	1.2622	0.3332	0.5121	0.3842	0.1537
ACU_LAT_4	ACU LAT	°C	0.7492	0.4512	0.8877	1.0722	0.7005	0.6306	0.3373	0.2253	0.5717	0.0581
ACU_LAT_5	ACU LAT	°C	0.0913	0.0761	0.2023	0.1015	0.1476	0.1699	0.1519	0.0832	0.1411	0.0537
ACU_LAT_6	ACU LAT	°C	0.3810	0.3825	0.2775	0.3461	0.3555	0.4633	0.3752	0.2251	0.5451	0.0394
ACU_LAT_7	ACU LAT	°C	0.3638	0.2223	1.1021	0.4128	0.6021	1.4397	0.2497	0.1465	0.2469	0.0239
ACU_LAT_8	ACU LAT	°C	1.6382	1.4211	1.5522	0.7752	1.2511	0.9122	0.5268	0.1261	0.7020	0.0534
ACU_EAT_0	ACU EAT	°C	0.2357	0.1566	0.4029	0.3978	0.3266	0.3411	0.0844	0.0859	0.1252	0.0281
ACU_EAT_1	ACU EAT	°C	0.1943	0.1281	0.2324	0.0611	0.2719	0.1871	0.0541	0.0715	0.0653	0.0309
ACU_EAT_2	ACU EAT	°C	0.1116	0.0870	0.1821	0.1715	0.1533	0.1993	0.2362	0.0736	0.0586	0.0375
ACU_EAT_3	ACU EAT	°C	0.1889	0.1350	0.3038	0.3127	0.2507	0.4656	0.2161	0.1261	0.0794	0.0371
ACU_EAT_4	ACU EAT	°C	0.1088	0.0821	0.1043	0.0446	0.0838	0.0605	0.0388	0.0294	0.0301	0.0126
ACU_EAT_5	ACU EAT	°C	0.2290	0.1353	0.2420	0.1660	0.2154	0.1979	0.0701	0.0684	0.0853	0.0239
ACU_EAT_6	ACU EAT	°C	0.2524	0.1770	0.2086	0.1051	0.1624	0.1009	0.0503	0.0466	0.0836	0.0169
ACU_EAT_7	ACU EAT	°C	0.3978	0.2604	0.3342	0.2004	0.3076	0.1198	0.0688	0.0353	0.0757	0.0189
ACU_EAT_8	ACU EAT	°C	0.6599	0.4761	0.6835	0.4668	0.4902	0.2428	0.1316	0.0350	0.1525	0.0201

Table 7: Detailed per-channel RMSE results corresponding to Table 2. The table reports RMSE in native physical units for all 46 output channels at $H = 10$ (50 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	Kalman- SS	Inform- er	TSMixer	TimesNet	LightTS	BOREAS	
cold_aisle_T_00	Cold-T	°C	1.1961	0.8887	1.0797	0.7960	1.0629	0.8273	1.4236	0.9721	1.3842	0.8239
cold_aisle_T_01	Cold-T	°C	1.3877	1.2748	1.3275	1.3089	1.4133	1.5616	1.5320	1.2718	1.3198	1.2517
cold_aisle_T_02	Cold-T	°C	1.2487	1.0746	1.2763	0.9637	1.2691	1.3245	1.2256	0.9274	1.0937	0.8168
cold_aisle_T_03	Cold-T	°C	0.7761	0.6332	0.7756	0.6408	0.7798	0.7455	0.9082	0.6676	0.7932	0.5661
cold_aisle_T_04	Cold-T	°C	0.5955	0.5418	0.6329	0.9968	0.9759	1.0763	0.7651	0.4326	0.5558	0.3309
cold_aisle_T_05	Cold-T	°C	0.6402	0.6095	0.7950	0.4768	0.6865	0.7179	0.4870	0.5327	0.4915	0.4078
cold_aisle_T_06	Cold-T	°C	1.4680	1.3633	1.2403	1.1334	1.2046	1.5904	1.5560	1.3328	1.1597	1.2893
cold_aisle_T_07	Cold-T	°C	1.3353	1.1992	1.0604	0.9482	1.0700	1.0745	1.5047	1.1509	1.1438	1.1315
cold_aisle_T_08	Cold-T	°C	0.7595	0.6681	0.8319	0.8982	0.8627	1.1536	0.9917	0.5758	0.5952	0.5750
cold_aisle_T_09	Cold-T	°C	0.9517	0.8100	0.7243	0.7334	0.8678	1.1631	0.9089	0.6914	0.5900	0.6603
cold_aisle_T_10	Cold-T	°C	1.1219	0.9280	0.8992	0.8586	0.6872	1.2211	0.6006	0.5761	0.8822	0.4764
cold_aisle_T_11	Cold-T	°C	1.6125	1.4953	1.3868	1.1334	1.3774	1.7190	1.1861	1.3009	1.1678	1.3303
hot_aisle_T_00	Hot-T	°C	0.6323	0.5895	0.6501	0.6277	0.7630	0.6219	0.4225	0.3311	0.3413	0.2811
hot_aisle_T_01	Hot-T	°C	0.6589	0.5399	0.6504	0.5798	0.6914	0.9686	0.4433	0.4335	0.5129	0.3157
cold_aisle_H_00	Cold-RH	%RH	3.6534	3.0025	3.9560	2.8279	3.7949	7.4379	4.8327	3.7316	4.6112	2.5161
cold_aisle_H_01	Cold-RH	%RH	4.2471	4.0360	4.3016	4.1948	4.7018	6.5951	4.4850	4.5888	4.1983	3.9722
cold_aisle_H_02	Cold-RH	%RH	3.9320	3.5389	3.7195	3.1949	3.7736	6.4971	4.5449	2.9816	3.4448	2.5175
cold_aisle_H_03	Cold-RH	%RH	2.1356	2.1174	2.7658	1.9574	2.6402	7.3452	4.0154	2.5147	2.6879	1.6799
cold_aisle_H_04	Cold-RH	%RH	2.8235	2.2269	2.2581	3.5885	3.5994	8.1151	3.9891	2.2253	2.4979	0.9712
cold_aisle_H_05	Cold-RH	%RH	2.1345	1.9000	2.4500	1.6103	2.1479	4.5490	3.4443	1.7685	2.1487	1.1545
cold_aisle_H_06	Cold-RH	%RH	4.4459	4.3258	4.0385	3.5763	3.9967	6.2168	4.6065	4.8489	4.4951	3.9413
cold_aisle_H_07	Cold-RH	%RH	4.2278	3.9739	3.8165	3.3948	3.8956	6.7671	4.2586	4.5300	4.4889	3.4241
cold_aisle_H_08	Cold-RH	%RH	2.1132	1.9937	3.1076	3.5172	3.1960	5.1657	3.0858	2.7924	2.5754	1.6294
cold_aisle_H_09	Cold-RH	%RH	2.5808	2.3459	2.3723	2.1697	2.9004	5.0702	2.9885	3.2327	2.5061	1.8110
cold_aisle_H_10	Cold-RH	%RH	2.6151	2.4187	2.6618	2.5064	2.0102	4.2063	2.9885	2.0270	2.5998	1.4341
cold_aisle_H_11	Cold-RH	%RH	5.2145	4.9629	5.1888	3.8329	5.0852	5.2914	4.9444	4.4389	4.1418	4.4542
hot_aisle_H_00	Hot-RH	%RH	1.5093	1.1113	1.0858	0.9585	1.2181	2.8014	1.1169	0.9518	0.9857	0.5783
hot_aisle_H_01	Hot-RH	%RH	1.4072	1.2242	1.3100	1.1303	1.2926	3.0250	1.4930	1.0465	1.1697	0.7466
ACU_LAT_0	ACU LAT	°C	1.4353	1.1340	1.3395	2.1607	1.3261	1.7612	1.3384	0.9274	1.2217	0.3534
ACU_LAT_1	ACU LAT	°C	1.0540	0.8123	0.9720	0.7467	1.0510	0.8642	0.6465	0.6946	0.7297	0.5826
ACU_LAT_2	ACU LAT	°C	0.6681	0.7094	0.9010	0.9393	0.9771	1.0413	0.8063	0.6307	0.8350	0.6179
ACU_LAT_3	ACU LAT	°C	1.5058	1.2534	1.3412	0.9473	1.2910	1.2798	0.7901	0.9835	0.9134	0.5182
ACU_LAT_4	ACU LAT	°C	1.1543	0.9428	1.1992	1.0533	1.1020	0.8954	0.7559	0.6346	1.1335	0.3496
ACU_LAT_5	ACU LAT	°C	0.4453	0.3879	0.5492	0.3425	0.5227	0.4580	0.5053	0.3960	0.4799	0.2917
ACU_LAT_6	ACU LAT	°C	0.8404	0.8855	0.6405	0.6315	0.6977	0.8007	0.7929	0.6156	0.9926	0.2666
ACU_LAT_7	ACU LAT	°C	0.8879	0.7303	1.4501	0.6793	1.1818	1.2717	0.7117	0.4939	0.6929	0.2055
ACU_LAT_8	ACU LAT	°C	1.7145	1.6406	1.7884	0.9813	1.7172	1.1970	0.9619	0.4878	1.1859	0.3081
ACU_EAT_0	ACU EAT	°C	0.7160	0.5794	0.8124	0.6785	0.6990	0.6275	0.3530	0.4362	0.4904	0.2502
ACU_EAT_1	ACU EAT	°C	0.6957	0.5299	0.6727	0.2733	0.7085	0.4709	0.3185	0.3856	0.3596	0.2326
ACU_EAT_2	ACU EAT	°C	0.5437	0.4370	0.6306	0.4615	0.6199	0.5381	0.6060	0.3868	0.3306	0.2356
ACU_EAT_3	ACU EAT	°C	0.7257	0.5910	0.7974	0.6122	0.7705	0.7702	0.6254	0.5394	0.3906	0.2498
ACU_EAT_4	ACU EAT	°C	0.5227	0.4575	0.5097	0.2462	0.4651	0.3207	0.2891	0.2681	0.2565	0.1615
ACU_EAT_5	ACU EAT	°C	0.7389	0.5890	0.6954	0.4384	0.6536	0.5261	0.3487	0.3881	0.3991	0.2123
ACU_EAT_6	ACU EAT	°C	0.7538	0.6534	0.7088	0.3677	0.6400	0.4309	0.3486	0.3113	0.4202	0.1896
ACU_EAT_7	ACU EAT	°C	0.9171	0.7868	0.8913	0.4942	0.8288	0.4606	0.3977	0.2779	0.4080	0.1958
ACU_EAT_8	ACU EAT	°C	1.1350	1.0377	1.1487	0.7498	1.0198	0.6012	0.4940	0.2671	0.5697	0.2018

Table 8: Detailed per-channel MAE results corresponding to Table 2. The table reports MAE in native physical units for all 46 output channels at $H = 10$ (50 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	Kalman- SS	Informer	TSMixer	TimesNet	LightTS	BOREAS	
cold_aisle_T_00	Cold-T	°C	0.9211	0.6315	0.7869	0.5429	0.7822	0.5567	1.1322	0.6917	0.9564	0.4920
cold_aisle_T_01	Cold-T	°C	1.0010	0.8255	0.8262	0.8612	0.9108	1.1637	1.1373	0.8172	0.8965	0.6851
cold_aisle_T_02	Cold-T	°C	0.9030	0.8281	0.9757	0.6964	0.9999	0.9957	1.0212	0.6319	0.8392	0.4669
cold_aisle_T_03	Cold-T	°C	0.5790	0.4398	0.5621	0.4217	0.5763	0.5705	0.6638	0.4327	0.5069	0.3337
cold_aisle_T_04	Cold-T	°C	0.4298	0.3760	0.4497	0.6216	0.6565	0.7251	0.4674	0.2798	0.3566	0.2069
cold_aisle_T_05	Cold-T	°C	0.4367	0.4452	0.6062	0.3320	0.5057	0.5774	0.3597	0.3612	0.3399	0.2685
cold_aisle_T_06	Cold-T	°C	1.0502	0.9735	0.9115	0.9091	0.9012	1.1957	1.1398	0.9349	0.7665	0.7912
cold_aisle_T_07	Cold-T	°C	0.9314	0.8253	0.7490	0.6959	0.7598	0.8222	1.0346	0.8039	0.7922	0.6917
cold_aisle_T_08	Cold-T	°C	0.5206	0.4411	0.6261	0.6727	0.6865	0.8684	0.7225	0.4028	0.4383	0.3430
cold_aisle_T_09	Cold-T	°C	0.6430	0.5268	0.4988	0.5671	0.6733	0.8563	0.5931	0.4626	0.3979	0.3671
cold_aisle_T_10	Cold-T	°C	0.7906	0.6204	0.5418	0.6355	0.4685	0.9535	0.4273	0.4131	0.6367	0.3237
cold_aisle_T_11	Cold-T	°C	1.0635	0.9354	0.8292	0.7642	0.8096	1.2215	0.6365	0.6911	0.6806	0.5650
hot_aisle_T_00	Hot-T	°C	0.4592	0.4305	0.4825	0.4565	0.5660	0.4288	0.3136	0.2118	0.2288	0.1670
hot_aisle_T_01	Hot-T	°C	0.4518	0.3901	0.4660	0.4484	0.5193	0.7503	0.2903	0.2961	0.3381	0.2112
cold_aisle_H_00	Cold-RH	%RH	2.6207	2.2998	2.8906	1.9861	2.7703	4.5490	3.6133	2.6196	3.2564	1.3715
cold_aisle_H_01	Cold-RH	%RH	2.8306	2.4375	2.7177	2.7774	3.0420	4.6778	3.0124	3.0814	2.8153	2.2611
cold_aisle_H_02	Cold-RH	%RH	2.9737	2.7767	2.8677	2.3015	2.9865	4.3656	3.6773	1.8964	2.5178	1.3793
cold_aisle_H_03	Cold-RH	%RH	1.4849	1.5428	1.9745	1.2717	1.9191	4.4480	2.8377	1.6485	1.9447	0.9083
cold_aisle_H_04	Cold-RH	%RH	2.0431	1.5425	1.6164	2.3271	2.5507	5.5127	2.8593	1.5257	1.6997	0.6011
cold_aisle_H_05	Cold-RH	%RH	1.6115	1.4320	1.8855	1.1883	1.6098	2.9545	2.4328	1.2846	1.6206	0.7511
cold_aisle_H_06	Cold-RH	%RH	3.3897	3.1022	2.9937	2.8754	2.9563	4.4633	3.4370	3.4279	3.5547	2.3048
cold_aisle_H_07	Cold-RH	%RH	3.0054	2.8301	2.8033	2.6537	2.8667	4.5832	2.9397	3.0569	3.3114	2.0847
cold_aisle_H_08	Cold-RH	%RH	1.4756	1.3669	2.3536	2.5760	2.6046	3.4271	2.2144	1.8750	1.8129	0.9668
cold_aisle_H_09	Cold-RH	%RH	1.8236	1.5891	1.6670	1.6858	2.2404	3.2967	1.9072	2.1906	1.6512	0.9925
cold_aisle_H_10	Cold-RH	%RH	1.7196	1.5926	1.6709	1.8686	1.4184	2.8414	2.0905	1.4834	1.9628	0.9439
cold_aisle_H_11	Cold-RH	%RH	3.2642	3.1023	3.1316	2.4900	3.1404	3.4545	3.1829	2.5475	2.8090	1.8069
hot_aisle_H_00	Hot-RH	%RH	1.0995	0.7840	0.8219	0.7598	0.9220	1.6558	0.7991	0.6256	0.7428	0.3830
hot_aisle_H_01	Hot-RH	%RH	0.9819	0.8534	0.9718	0.7749	0.9525	2.0755	1.0906	0.7172	0.8199	0.5149
ACU_LAT_0	ACU LAT	°C	1.1924	0.9504	1.1059	1.7010	1.0528	1.4384	0.9647	0.6269	0.8574	0.1680
ACU_LAT_1	ACU LAT	°C	0.7775	0.5042	0.6548	0.5035	0.7529	0.6265	0.4074	0.3986	0.3482	0.2478
ACU_LAT_2	ACU LAT	°C	0.3843	0.3760	0.6113	0.6499	0.6705	0.7221	0.5504	0.3369	0.4700	0.2302
ACU_LAT_3	ACU LAT	°C	1.0193	0.9003	0.8568	0.7057	0.7316	0.9333	0.5392	0.7004	0.6121	0.2492
ACU_LAT_4	ACU LAT	°C	0.9439	0.7493	0.9240	0.8068	0.8376	0.6884	0.5227	0.4224	0.8631	0.2213
ACU_LAT_5	ACU LAT	°C	0.3160	0.2547	0.4156	0.2327	0.3862	0.3244	0.3554	0.2727	0.3372	0.1616
ACU_LAT_6	ACU LAT	°C	0.6347	0.6554	0.4926	0.4874	0.5502	0.5982	0.5646	0.4012	0.7055	0.1740
ACU_LAT_7	ACU LAT	°C	0.5210	0.4364	0.9930	0.5245	0.8211	0.9616	0.5139	0.3426	0.4717	0.1106
ACU_LAT_8	ACU LAT	°C	1.0591	0.9455	1.1904	0.7780	1.1748	0.8393	0.7059	0.3508	0.8392	0.1881
ACU_EAT_0	ACU EAT	°C	0.5523	0.4448	0.6515	0.5710	0.6019	0.5060	0.2484	0.2850	0.3705	0.1514
ACU_EAT_1	ACU EAT	°C	0.4805	0.3785	0.4873	0.1866	0.5122	0.3371	0.1979	0.2520	0.2299	0.1311
ACU_EAT_2	ACU EAT	°C	0.3646	0.2703	0.4375	0.3656	0.3863	0.3807	0.4302	0.2203	0.2137	0.1291
ACU_EAT_3	ACU EAT	°C	0.4892	0.4019	0.5772	0.4631	0.4953	0.5404	0.4165	0.3474	0.2679	0.1392
ACU_EAT_4	ACU EAT	°C	0.3973	0.3421	0.3790	0.1868	0.3746	0.2602	0.2086	0.1947	0.2051	0.1167
ACU_EAT_5	ACU EAT	°C	0.5511	0.4255	0.5130	0.2888	0.5005	0.3666	0.2447	0.2808	0.2873	0.1268
ACU_EAT_6	ACU EAT	°C	0.4867	0.4239	0.4457	0.2491	0.4420	0.3074	0.2374	0.2138	0.2993	0.1243
ACU_EAT_7	ACU EAT	°C	0.5670	0.4625	0.5394	0.3335	0.5398	0.3277	0.2738	0.1951	0.2900	0.1227
ACU_EAT_8	ACU EAT	°C	0.6655	0.5842	0.6854	0.4798	0.6004	0.4255	0.3398	0.1845	0.3827	0.1356

Table 9: Detailed per-channel MSE results corresponding to Table 2. The table reports MSE in squared native units for all 46 output channels at $H = 10$ (50 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	DMDc	Subspace- SS	Kalman- SS	Informer	TSMixer	TimesNet	LightTS	BOREAS
cold_aisle_T_00	Cold-T	°C	1.4306	0.7898	1.1657	0.6337	1.1297	0.6845	2.0267	0.9449	1.9159	0.6788
cold_aisle_T_01	Cold-T	°C	1.9258	1.6252	1.7621	1.7132	1.9974	2.4387	2.3470	1.6174	1.7420	1.5667
cold_aisle_T_02	Cold-T	°C	1.5591	1.1547	1.6290	0.9287	1.6106	1.7543	1.5022	0.8600	1.1963	0.6671
cold_aisle_T_03	Cold-T	°C	0.6023	0.4010	0.6016	0.4106	0.6081	0.5557	0.8247	0.4457	0.6291	0.3205
cold_aisle_T_04	Cold-T	°C	0.3546	0.2936	0.4006	0.9936	0.9523	1.1584	0.5854	0.1871	0.3090	0.1095
cold_aisle_T_05	Cold-T	°C	0.4098	0.3715	0.6321	0.2273	0.4712	0.5154	0.2371	0.2838	0.2416	0.1663
cold_aisle_T_06	Cold-T	°C	2.1550	1.8586	1.5383	1.2845	1.4512	2.5294	2.4211	1.7763	1.3450	1.6624
cold_aisle_T_07	Cold-T	°C	1.7830	1.4380	1.1245	0.8990	1.1450	1.1546	2.2642	1.3245	1.3084	1.2803
cold_aisle_T_08	Cold-T	°C	0.5769	0.4463	0.6921	0.8067	0.7443	1.3307	0.9834	0.3315	0.3543	0.3306
cold_aisle_T_09	Cold-T	°C	0.9056	0.6561	0.5246	0.5379	0.7531	1.3529	0.8261	0.4781	0.3481	0.4360
cold_aisle_T_10	Cold-T	°C	1.2587	0.8612	0.8086	0.7372	0.4722	1.4911	0.3607	0.3319	0.7782	0.2269
cold_aisle_T_11	Cold-T	°C	2.6003	2.2358	1.9233	1.2847	1.8972	2.9550	1.4069	1.6923	1.3637	1.7697
hot_aisle_T_00	Hot-T	°C	0.3998	0.3475	0.4226	0.3940	0.5821	0.3868	0.1785	0.1097	0.1165	0.0790
hot_aisle_T_01	Hot-T	°C	0.4342	0.2915	0.4230	0.3361	0.4780	0.9381	0.1965	0.1879	0.2630	0.0996
cold_aisle_H_00	Cold-RH	%RH	13.3474	9.0151	15.6503	7.9972	14.4013	55.3226	23.3548	13.9249	21.2628	6.3307
cold_aisle_H_01	Cold-RH	%RH	18.0376	16.2895	18.5036	17.5960	22.1065	43.4953	20.1154	21.0574	17.6261	15.7784
cold_aisle_H_02	Cold-RH	%RH	15.4610	12.5236	13.8350	10.2076	14.2402	42.2128	20.6564	8.8898	11.8667	6.3378
cold_aisle_H_03	Cold-RH	%RH	4.5608	4.4833	7.6495	3.8313	6.9707	53.9523	16.1238	6.3238	7.2249	2.8219
cold_aisle_H_04	Cold-RH	%RH	7.9719	4.9589	5.0992	12.8774	12.9556	65.8547	15.9132	4.9520	6.2393	0.9433
cold_aisle_H_05	Cold-RH	%RH	4.5561	3.6100	6.0026	2.5931	4.6136	20.6932	11.8632	3.1275	4.6167	1.3328
cold_aisle_H_06	Cold-RH	%RH	19.7663	18.7122	16.3097	12.7896	15.9735	38.6489	21.2201	23.5120	20.2057	15.5336
cold_aisle_H_07	Cold-RH	%RH	17.8745	15.7920	14.5660	11.5248	15.1755	45.7936	18.1360	20.5213	20.1502	11.7247
cold_aisle_H_08	Cold-RH	%RH	4.4657	3.9749	9.6571	12.3706	10.2144	26.6840	9.5223	7.7975	6.6326	2.6549
cold_aisle_H_09	Cold-RH	%RH	6.6605	5.5031	5.6276	4.7074	8.4121	25.7069	8.9312	10.4504	6.2806	3.2796
cold_aisle_H_10	Cold-RH	%RH	6.8386	5.8501	7.0852	6.2822	4.0409	17.6927	8.9311	4.1089	6.7592	2.0566
cold_aisle_H_11	Cold-RH	%RH	27.1909	24.6299	26.9233	14.6909	25.8591	27.9994	24.4474	19.7035	17.1544	19.8395
hot_aisle_H_00	Hot-RH	%RH	2.2780	1.2351	1.1789	0.9188	1.4837	7.8478	1.2474	0.9059	0.9717	0.3344
hot_aisle_H_01	Hot-RH	%RH	1.9803	1.4987	1.7160	1.2775	1.6709	9.1507	2.2292	1.0951	1.3683	0.5574
ACU_LAT_0	ACU LAT	°C	2.0600	1.2861	1.7944	4.6685	1.7585	3.1017	1.7913	0.8601	1.4927	0.1249
ACU_LAT_1	ACU LAT	°C	1.1110	0.6598	0.9448	0.5576	1.1047	0.7468	0.4180	0.4825	0.5324	0.3394
ACU_LAT_2	ACU LAT	°C	0.4464	0.5032	0.8117	0.8823	0.9547	1.0843	0.6501	0.3978	0.6973	0.3818
ACU_LAT_3	ACU LAT	°C	2.2675	1.5709	1.7988	0.8975	1.6667	1.6379	0.6242	0.9673	0.8344	0.2685
ACU_LAT_4	ACU LAT	°C	1.3325	0.8889	1.4381	1.1095	1.2143	0.8017	0.5715	0.4027	1.2849	0.1222
ACU_LAT_5	ACU LAT	°C	0.1983	0.1505	0.3016	0.1173	0.2732	0.2097	0.2554	0.1568	0.2303	0.0851
ACU_LAT_6	ACU LAT	°C	0.7062	0.7841	0.4103	0.3987	0.4868	0.6412	0.6287	0.3790	0.9853	0.0711
ACU_LAT_7	ACU LAT	°C	0.7884	0.5334	2.1029	0.4615	1.3967	1.6171	0.5066	0.2440	0.4800	0.0422
ACU_LAT_8	ACU LAT	°C	2.9396	2.6916	3.1984	0.9630	2.9488	1.4329	0.9253	0.2379	1.4064	0.0949
ACU_EAT_0	ACU EAT	°C	0.5126	0.3357	0.6599	0.4603	0.4886	0.3937	0.1246	0.1902	0.2405	0.0626
ACU_EAT_1	ACU EAT	°C	0.4840	0.2808	0.4526	0.0747	0.5019	0.2218	0.1014	0.1487	0.1293	0.0541
ACU_EAT_2	ACU EAT	°C	0.2956	0.1910	0.3976	0.2129	0.3843	0.2895	0.3672	0.1496	0.1093	0.0555
ACU_EAT_3	ACU EAT	°C	0.5267	0.3493	0.6359	0.3748	0.5937	0.5932	0.3912	0.2910	0.1526	0.0624
ACU_EAT_4	ACU EAT	°C	0.2732	0.2093	0.2598	0.0606	0.2163	0.1029	0.0836	0.0719	0.0658	0.0261
ACU_EAT_5	ACU EAT	°C	0.5460	0.3470	0.4836	0.1922	0.4272	0.2768	0.1216	0.1506	0.1593	0.0451
ACU_EAT_6	ACU EAT	°C	0.5682	0.4269	0.5025	0.1352	0.4096	0.1857	0.1216	0.0969	0.1766	0.0359
ACU_EAT_7	ACU EAT	°C	0.8411	0.6191	0.7945	0.2443	0.6869	0.2122	0.1582	0.0772	0.1664	0.0383
ACU_EAT_8	ACU EAT	°C	1.2881	1.0768	1.3195	0.5622	1.0400	0.3615	0.2440	0.0713	0.3245	0.0407

Table 10: Detailed per-channel RMSE results corresponding to Table 2. The table reports RMSE in native physical units for all 46 output channels at $H = 15$ (75 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	Kalman- SS	Inform- er	TSMixer	TimesNet	LightTS	BOREAS	
cold_aisle_T_00	Cold-T	°C	1.3053	1.0446	1.2324	0.7965	1.2850	0.9154	1.6742	1.1530	1.6906	0.8924
cold_aisle_T_01	Cold-T	°C	1.4966	1.3672	1.4191	1.3129	1.4680	1.6203	1.7492	1.3356	1.4754	1.2659
cold_aisle_T_02	Cold-T	°C	1.3638	1.2377	1.4383	0.9961	1.4345	1.4096	1.4314	1.0130	1.2787	0.8703
cold_aisle_T_03	Cold-T	°C	0.8829	0.7172	0.8586	0.6742	0.8803	0.8401	1.0557	0.7687	1.0032	0.6109
cold_aisle_T_04	Cold-T	°C	0.6813	0.6426	0.7183	1.0354	0.9973	1.1447	0.8912	0.5188	0.6663	0.3837
cold_aisle_T_05	Cold-T	°C	0.7210	0.7110	0.9244	0.5019	0.8430	0.7644	0.5573	0.6575	0.5921	0.4602
cold_aisle_T_06	Cold-T	°C	1.5732	1.4417	1.2423	1.1369	1.2115	1.6493	1.7849	1.3983	1.2432	1.3321
cold_aisle_T_07	Cold-T	°C	1.3810	1.2403	1.0643	0.9626	1.0716	1.1219	1.7705	1.1985	1.3356	1.2207
cold_aisle_T_08	Cold-T	°C	0.8335	0.7337	0.8852	0.9096	0.8597	1.1821	1.1241	0.6173	0.6720	0.6224
cold_aisle_T_09	Cold-T	°C	1.0324	0.9055	0.7688	0.7494	0.8758	1.2164	1.0471	0.7410	0.6513	0.7135
cold_aisle_T_10	Cold-T	°C	1.2702	1.1317	1.0458	0.8731	0.8042	1.2784	0.7017	0.6819	1.1149	0.5376
cold_aisle_T_11	Cold-T	°C	1.7884	1.6566	1.5014	1.1730	1.5116	1.8974	1.2866	1.4152	1.2813	1.3890
hot_aisle_T_00	Hot-T	°C	0.7255	0.6925	0.7675	0.6584	0.8320	0.6393	0.4851	0.3713	0.3953	0.2934
hot_aisle_T_01	Hot-T	°C	0.7509	0.6419	0.7688	0.5862	0.7662	1.0064	0.5383	0.5379	0.6409	0.3388
cold_aisle_H_00	Cold-RH	%RH	4.2808	3.5240	4.6610	2.7963	4.7284	8.4858	6.0635	4.5331	5.8365	2.7202
cold_aisle_H_01	Cold-RH	%RH	4.5309	4.2478	4.5826	4.1744	4.8912	7.1413	5.1527	5.0065	4.5174	4.0588
cold_aisle_H_02	Cold-RH	%RH	4.5000	4.1828	4.3855	3.3058	4.4521	6.9884	5.4524	3.3567	4.1971	2.6522
cold_aisle_H_03	Cold-RH	%RH	2.5102	2.3259	3.1281	2.0480	3.0388	8.2838	4.9951	3.1444	3.4171	1.8195
cold_aisle_H_04	Cold-RH	%RH	3.3569	2.7173	2.6718	3.7042	3.6778	8.6984	4.8503	2.8505	3.2839	1.1168
cold_aisle_H_05	Cold-RH	%RH	2.6433	2.2978	2.9184	1.7128	2.7129	5.0019	3.9867	2.0618	2.6676	1.2843
cold_aisle_H_06	Cold-RH	%RH	4.6621	4.5441	3.9681	3.5782	3.9461	6.7817	5.2181	5.3672	5.0741	3.9931
cold_aisle_H_07	Cold-RH	%RH	4.2923	4.1468	3.7425	3.4336	3.8404	7.4493	5.0659	5.0299	5.4965	3.6453
cold_aisle_H_08	Cold-RH	%RH	2.2627	2.1539	3.2861	3.5853	3.1224	5.6593	3.6387	3.4892	3.1537	1.7230
cold_aisle_H_09	Cold-RH	%RH	2.7437	2.5797	2.5022	2.2788	2.9059	5.5262	3.5254	4.1035	3.0195	1.9212
cold_aisle_H_10	Cold-RH	%RH	2.9779	2.8510	3.1213	2.5597	2.3781	4.5774	3.4827	2.3716	3.2637	1.6338
cold_aisle_H_11	Cold-RH	%RH	5.7215	5.4829	5.6675	3.9900	5.6674	5.6487	5.6311	4.9559	4.7703	4.6585
hot_aisle_H_00	Hot-RH	%RH	1.7410	1.3380	1.3131	1.0173	1.3825	3.0782	1.3153	1.2319	1.2303	0.6592
hot_aisle_H_01	Hot-RH	%RH	1.5866	1.4325	1.5707	1.1669	1.4999	3.3191	1.7408	1.2987	1.4673	0.8349
ACU_LAT_0	ACU LAT	°C	1.5619	1.2626	1.4040	2.1737	1.4463	1.9688	1.5779	1.1427	1.4871	0.4587
ACU_LAT_1	ACU LAT	°C	1.2462	0.9033	1.0522	0.7649	1.1558	0.8803	0.7199	0.8141	0.7877	0.6400
ACU_LAT_2	ACU LAT	°C	0.8213	0.8084	1.0295	0.9764	1.1385	1.0995	0.9036	0.7089	0.9748	0.6510
ACU_LAT_3	ACU LAT	°C	1.9371	1.5391	1.6704	1.0127	1.7185	1.3864	0.9937	1.2333	1.2589	0.6386
ACU_LAT_4	ACU LAT	°C	1.3756	1.0874	1.3787	1.0762	1.3386	0.9609	0.9216	0.7712	1.4437	0.4581
ACU_LAT_5	ACU LAT	°C	0.5650	0.4672	0.6352	0.3650	0.6615	0.5006	0.6028	0.5292	0.5670	0.3378
ACU_LAT_6	ACU LAT	°C	0.9579	1.0280	0.7311	0.6669	0.8202	0.8775	0.9374	0.6993	1.1846	0.3286
ACU_LAT_7	ACU LAT	°C	1.0912	0.9179	1.7375	0.7343	1.5130	1.3439	0.8871	0.5865	0.8562	0.2505
ACU_LAT_8	ACU LAT	°C	1.9535	1.8594	2.1890	1.0766	2.2126	1.3535	1.1449	0.6354	1.4603	0.3818
ACU_EAT_0	ACU EAT	°C	0.9037	0.7270	0.9185	0.7221	0.8340	0.6642	0.4167	0.5865	0.6194	0.3385
ACU_EAT_1	ACU EAT	°C	0.9330	0.6923	0.8476	0.2989	0.9185	0.5168	0.4140	0.5056	0.4780	0.2972
ACU_EAT_2	ACU EAT	°C	0.7574	0.5920	0.8301	0.5073	0.8674	0.6112	0.7291	0.5144	0.4400	0.2863
ACU_EAT_3	ACU EAT	°C	1.0096	0.8187	1.0472	0.6688	1.0835	0.8528	0.7749	0.7256	0.5402	0.3119
ACU_EAT_4	ACU EAT	°C	0.6550	0.5732	0.6473	0.2751	0.6302	0.3806	0.3746	0.3603	0.3390	0.2110
ACU_EAT_5	ACU EAT	°C	0.9244	0.7510	0.8645	0.4647	0.8422	0.5917	0.4372	0.5027	0.5003	0.2687
ACU_EAT_6	ACU EAT	°C	0.9081	0.8060	0.8956	0.4019	0.8396	0.5174	0.4541	0.3823	0.5315	0.2472
ACU_EAT_7	ACU EAT	°C	1.0860	0.9663	1.1184	0.5341	1.0585	0.5505	0.5121	0.3605	0.5243	0.2510
ACU_EAT_8	ACU EAT	°C	1.3162	1.2500	1.3914	0.8057	1.2836	0.6887	0.5998	0.3488	0.7253	0.2551

Table 11: Detailed per-channel MAE results corresponding to Table 2. The table reports MAE in native physical units for all 46 output channels at $H = 15$ (75 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	Subspace- DMDc	Kalman- SS	Inform- er	TSMixer	TimesNet	LightTS	BOREAS	
cold_aisle_T_00	Cold-T	°C	0.9872	0.7418	0.9018	0.5394	0.9464	0.6443	1.3359	0.8234	1.1784	0.5609
cold_aisle_T_01	Cold-T	°C	1.0966	0.9264	0.9240	0.8710	0.9790	1.2233	1.3592	0.8693	1.0309	0.7125
cold_aisle_T_02	Cold-T	°C	0.9921	0.9670	1.1169	0.7376	1.1420	1.0586	1.2050	0.7235	0.9963	0.5306
cold_aisle_T_03	Cold-T	°C	0.6535	0.5023	0.6363	0.4444	0.6663	0.6531	0.7780	0.5068	0.6388	0.3811
cold_aisle_T_04	Cold-T	°C	0.4865	0.4422	0.5086	0.6481	0.6819	0.7878	0.5466	0.3476	0.4328	0.2479
cold_aisle_T_05	Cold-T	°C	0.4957	0.5227	0.7202	0.3492	0.6242	0.6277	0.4121	0.4425	0.4113	0.3086
cold_aisle_T_06	Cold-T	°C	1.1389	1.0608	0.9387	0.9140	0.9254	1.2476	1.2794	0.9941	0.8439	0.8354
cold_aisle_T_07	Cold-T	°C	0.9772	0.8740	0.7485	0.6989	0.7497	0.8516	1.2263	0.8501	0.9283	0.7584
cold_aisle_T_08	Cold-T	°C	0.5821	0.4957	0.6743	0.6822	0.6870	0.8858	0.8048	0.4405	0.4971	0.3814
cold_aisle_T_09	Cold-T	°C	0.7135	0.6050	0.5356	0.5747	0.6701	0.8818	0.6904	0.5061	0.4555	0.4165
cold_aisle_T_10	Cold-T	°C	0.9058	0.7759	0.6613	0.6476	0.5535	1.0045	0.5018	0.4821	0.8077	0.3716
cold_aisle_T_11	Cold-T	°C	1.2285	1.0954	0.9724	0.8159	0.9546	1.3661	0.7162	0.7835	0.7628	0.6167
hot_aisle_T_00	Hot-T	°C	0.5211	0.5020	0.5630	0.4801	0.6177	0.4467	0.3601	0.2414	0.2694	0.1786
hot_aisle_T_01	Hot-T	°C	0.5149	0.4583	0.5584	0.4478	0.5857	0.7794	0.3613	0.3646	0.4284	0.2340
cold_aisle_H_00	Cold-RH	%RH	2.9851	2.6357	3.3545	1.9560	3.3853	5.2461	4.6316	3.1426	4.2411	1.5621
cold_aisle_H_01	Cold-RH	%RH	3.0973	2.6974	3.0564	2.7886	3.3096	5.1965	3.6855	3.4939	3.1641	2.3945
cold_aisle_H_02	Cold-RH	%RH	3.4862	3.3730	3.4448	2.4200	3.5737	4.6716	4.3816	2.1572	3.0768	1.5462
cold_aisle_H_03	Cold-RH	%RH	1.7005	1.7328	2.2566	1.3488	2.2584	5.0929	3.5650	2.0628	2.4466	1.0178
cold_aisle_H_04	Cold-RH	%RH	2.4936	1.9278	1.9344	2.3975	2.6651	5.9380	3.4327	1.9783	2.2350	0.6990
cold_aisle_H_05	Cold-RH	%RH	2.0203	1.7596	2.2880	1.2724	2.0564	3.2872	2.8170	1.5215	1.9747	0.8399
cold_aisle_H_06	Cold-RH	%RH	3.6561	3.4002	2.9952	2.8867	2.9726	4.9080	3.8599	3.8584	4.0656	2.3780
cold_aisle_H_07	Cold-RH	%RH	3.0915	3.0499	2.7794	2.6636	2.8342	5.1306	3.4211	3.4227	4.0520	2.2440
cold_aisle_H_08	Cold-RH	%RH	1.6066	1.5257	2.5181	2.6315	2.5607	3.7708	2.5609	2.3070	2.1883	1.0546
cold_aisle_H_09	Cold-RH	%RH	2.0062	1.8125	1.7874	1.7531	2.2307	3.6302	2.2439	2.7410	1.9715	1.0915
cold_aisle_H_10	Cold-RH	%RH	2.0443	1.9633	2.0521	1.9122	1.7266	3.1461	2.4068	1.6991	2.4791	1.0890
cold_aisle_H_11	Cold-RH	%RH	3.7853	3.6363	3.7004	2.7053	3.7588	3.7911	3.7136	2.9823	3.3111	1.9500
hot_aisle_H_00	Hot-RH	%RH	1.2620	0.9342	0.9608	0.7995	1.0217	1.8305	0.9245	0.8416	0.9068	0.4479
hot_aisle_H_01	Hot-RH	%RH	1.1304	1.0124	1.1637	0.8006	1.0945	2.2768	1.2548	0.9239	1.0025	0.5806
ACU_LAT_0	ACU LAT	°C	1.2799	1.0435	1.1779	1.7160	1.1618	1.6335	1.1472	0.7711	1.0533	0.2188
ACU_LAT_1	ACU LAT	°C	0.9400	0.6047	0.7385	0.5237	0.8465	0.6407	0.4631	0.4824	0.4111	0.2982
ACU_LAT_2	ACU LAT	°C	0.4959	0.4717	0.7224	0.6754	0.7758	0.7635	0.6355	0.4159	0.5825	0.2560
ACU_LAT_3	ACU LAT	°C	1.2679	1.1072	1.0817	0.7398	0.9734	1.0320	0.6720	0.8891	0.8368	0.3099
ACU_LAT_4	ACU LAT	°C	1.0849	0.8575	1.0392	0.8298	1.0072	0.7209	0.6310	0.4976	1.0754	0.2915
ACU_LAT_5	ACU LAT	°C	0.4003	0.3176	0.4808	0.2500	0.4780	0.3585	0.4232	0.3607	0.4068	0.2062
ACU_LAT_6	ACU LAT	°C	0.7173	0.7615	0.5645	0.5190	0.6395	0.6690	0.6579	0.4482	0.8374	0.2182
ACU_LAT_7	ACU LAT	°C	0.6666	0.5675	1.2119	0.5827	1.0584	1.0110	0.6469	0.4056	0.5873	0.1404
ACU_LAT_8	ACU LAT	°C	1.2236	1.1109	1.4767	0.8445	1.5215	0.9605	0.8530	0.4473	1.0304	0.2420
ACU_EAT_0	ACU EAT	°C	0.6974	0.5653	0.7392	0.6055	0.7140	0.5407	0.2907	0.3717	0.4700	0.2007
ACU_EAT_1	ACU EAT	°C	0.6459	0.4978	0.6267	0.2036	0.6611	0.3744	0.2511	0.3253	0.3043	0.1673
ACU_EAT_2	ACU EAT	°C	0.4892	0.3707	0.5648	0.4028	0.5338	0.4432	0.5232	0.2877	0.2779	0.1618
ACU_EAT_3	ACU EAT	°C	0.6538	0.5544	0.7403	0.5039	0.6548	0.6282	0.5293	0.4606	0.3606	0.1776
ACU_EAT_4	ACU EAT	°C	0.4961	0.4301	0.4781	0.2066	0.4934	0.3055	0.2624	0.2554	0.2707	0.1535
ACU_EAT_5	ACU EAT	°C	0.6846	0.5449	0.6264	0.3104	0.6329	0.4236	0.3038	0.3681	0.3589	0.1602
ACU_EAT_6	ACU EAT	°C	0.5948	0.5292	0.5743	0.2742	0.5823	0.3826	0.3094	0.2637	0.3835	0.1629
ACU_EAT_7	ACU EAT	°C	0.6704	0.5725	0.6920	0.3582	0.6972	0.4049	0.3551	0.2568	0.3764	0.1559
ACU_EAT_8	ACU EAT	°C	0.7622	0.7051	0.8533	0.5103	0.7862	0.5010	0.4196	0.2432	0.4905	0.1729

Table 12: Detailed per-channel MSE results corresponding to Table 2. The table reports MSE in squared native units for all 46 output channels at $H = 15$ (75 min), using the same 10,985 test windows and the updated BOREAS model with encoder width $d_{\text{model}} = 64$. Lower is better.

Channel	Group	Unit	OLS- ARX	Ridge- ARX	DMDc	Subspace- SS	Kalman- SS	Informer	TSMixer	TimesNet	LightTS	BOREAS
cold_aisle_T_00	Cold-T	°C	1.7038	1.0911	1.5187	0.6344	1.6512	0.8380	2.8030	1.3294	2.8582	0.7964
cold_aisle_T_01	Cold-T	°C	2.2398	1.8692	2.0138	1.7238	2.1550	2.6252	3.0598	1.7838	2.1767	1.6025
cold_aisle_T_02	Cold-T	°C	1.8600	1.5318	2.0688	0.9922	2.0577	1.9870	2.0489	1.0261	1.6352	0.7575
cold_aisle_T_03	Cold-T	°C	0.7796	0.5144	0.7373	0.4545	0.7749	0.7058	1.1144	0.5910	1.0064	0.3732
cold_aisle_T_04	Cold-T	°C	0.4642	0.4130	0.5160	1.0721	0.9947	1.3103	0.7942	0.2692	0.4439	0.1472
cold_aisle_T_05	Cold-T	°C	0.5198	0.5055	0.8546	0.2519	0.7107	0.5842	0.3106	0.4324	0.3505	0.2118
cold_aisle_T_06	Cold-T	°C	2.4748	2.0784	1.5434	1.2926	1.4677	2.7201	3.1859	1.9552	1.5454	1.7745
cold_aisle_T_07	Cold-T	°C	1.9073	1.5383	1.1327	0.9266	1.1484	1.2586	3.1348	1.4363	1.7838	1.4900
cold_aisle_T_08	Cold-T	°C	0.6946	0.5383	0.7836	0.8274	0.7391	1.3974	1.2636	0.3811	0.4515	0.3873
cold_aisle_T_09	Cold-T	°C	1.0659	0.8199	0.5910	0.5615	0.7670	1.4796	1.0964	0.5490	0.4242	0.5091
cold_aisle_T_10	Cold-T	°C	1.6135	1.2807	1.0936	0.7622	0.6467	1.6342	0.4924	0.4650	1.2430	0.2890
cold_aisle_T_11	Cold-T	°C	3.1984	2.7445	2.2541	1.3760	2.2850	3.6003	1.6554	2.0028	1.6417	1.9293
hot_aisle_T_00	Hot-T	°C	0.5264	0.4795	0.5891	0.4334	0.6921	0.4087	0.2353	0.1379	0.1562	0.0861
hot_aisle_T_01	Hot-T	°C	0.5639	0.4120	0.5911	0.3437	0.5871	1.0129	0.2898	0.2893	0.4108	0.1148
cold_aisle_H_00	Cold-RH	%RH	18.3250	12.4189	21.7247	7.8193	22.3579	72.0086	36.7657	20.5491	34.0642	7.3998
cold_aisle_H_01	Cold-RH	%RH	20.5287	18.0439	21.0003	17.4252	23.9234	50.9980	26.5506	25.0647	20.4072	16.4739
cold_aisle_H_02	Cold-RH	%RH	20.2500	17.4955	19.2326	10.9282	19.8212	48.8376	29.7282	11.2672	17.6157	7.0342
cold_aisle_H_03	Cold-RH	%RH	6.3009	5.4096	9.7852	4.1941	9.2343	68.6210	24.9508	9.8873	11.6769	3.3105
cold_aisle_H_04	Cold-RH	%RH	11.2690	7.3835	7.1383	13.7213	13.5265	75.6621	23.5251	8.1252	10.7838	1.2471
cold_aisle_H_05	Cold-RH	%RH	6.9869	5.2797	8.5169	2.9338	7.3598	25.0187	15.8941	4.2509	7.1160	1.6494
cold_aisle_H_06	Cold-RH	%RH	21.7347	20.6492	15.7454	12.8032	15.5718	45.9912	27.2285	28.8069	25.7467	15.9452
cold_aisle_H_07	Cold-RH	%RH	18.4242	17.1963	14.0065	11.7895	14.7489	55.4923	25.6635	25.2995	30.2111	13.2879
cold_aisle_H_08	Cold-RH	%RH	5.1200	4.6392	10.7982	12.8544	9.7494	32.0274	13.2399	12.1746	9.9456	2.9689
cold_aisle_H_09	Cold-RH	%RH	7.5280	6.6547	6.2608	5.1928	8.4440	30.5392	12.4283	16.8384	9.1176	3.6912
cold_aisle_H_10	Cold-RH	%RH	8.8676	8.1283	9.7427	6.5523	5.6553	20.9524	12.1292	5.6245	10.6518	2.6694
cold_aisle_H_11	Cold-RH	%RH	32.7359	30.0622	32.1202	15.9198	32.1195	31.9083	31.7088	24.5611	22.7559	21.7016
hot_aisle_H_00	Hot-RH	%RH	3.0312	1.7903	1.7241	1.0350	1.9114	9.4754	1.7301	1.5176	1.5135	0.4345
hot_aisle_H_01	Hot-RH	%RH	2.5174	2.0519	2.4672	1.3615	2.2496	11.0165	3.0303	1.6865	2.1530	0.6971
ACU_LAT_0	ACU LAT	°C	2.4394	1.5942	1.9714	4.7249	2.0918	3.8762	2.4897	1.3057	2.2115	0.2104
ACU_LAT_1	ACU LAT	°C	1.5529	0.8159	1.1071	0.5851	1.3358	0.7749	0.5183	0.6627	0.6205	0.4096
ACU_LAT_2	ACU LAT	°C	0.6745	0.6535	1.0598	0.9533	1.2962	1.2089	0.8165	0.5025	0.9502	0.4238
ACU_LAT_3	ACU LAT	°C	3.7525	2.3687	2.7902	1.0256	2.9534	1.9222	0.9874	1.5210	1.5848	0.4078
ACU_LAT_4	ACU LAT	°C	1.8922	1.1824	1.9008	1.1582	1.7918	0.9233	0.8493	0.5947	2.0842	0.2099
ACU_LAT_5	ACU LAT	°C	0.3192	0.2183	0.4035	0.1332	0.4376	0.2506	0.3634	0.2800	0.3215	0.1141
ACU_LAT_6	ACU LAT	°C	0.9175	1.0567	0.5346	0.4448	0.6728	0.7700	0.8788	0.4890	1.4032	0.1080
ACU_LAT_7	ACU LAT	°C	1.1906	0.8425	3.0188	0.5391	2.2891	1.8060	0.7869	0.3440	0.7331	0.0627
ACU_LAT_8	ACU LAT	°C	3.8160	3.4573	4.7917	1.1591	4.8955	1.8319	1.3108	0.4037	2.1325	0.1458
ACU_EAT_0	ACU EAT	°C	0.8166	0.5285	0.8436	0.5215	0.6955	0.4412	0.1737	0.3440	0.3836	0.1146
ACU_EAT_1	ACU EAT	°C	0.8704	0.4793	0.7185	0.0893	0.8436	0.2671	0.1714	0.2556	0.2285	0.0883
ACU_EAT_2	ACU EAT	°C	0.5737	0.3505	0.6890	0.2574	0.7524	0.3735	0.5315	0.2646	0.1936	0.0819
ACU_EAT_3	ACU EAT	°C	1.0193	0.6703	1.0967	0.4474	1.1739	0.7272	0.6004	0.5265	0.2918	0.0973
ACU_EAT_4	ACU EAT	°C	0.4290	0.3286	0.4191	0.0757	0.3971	0.1448	0.1403	0.1298	0.1149	0.0445
ACU_EAT_5	ACU EAT	°C	0.8544	0.5639	0.7473	0.2160	0.7093	0.3502	0.1912	0.2527	0.2503	0.0722
ACU_EAT_6	ACU EAT	°C	0.8247	0.6496	0.8022	0.1615	0.7050	0.2678	0.2062	0.1462	0.2825	0.0611
ACU_EAT_7	ACU EAT	°C	1.1794	0.9337	1.2509	0.2853	1.1204	0.3031	0.2623	0.1300	0.2749	0.0630
ACU_EAT_8	ACU EAT	°C	1.7324	1.5625	1.9360	0.6492	1.6476	0.4744	0.3597	0.1217	0.5260	0.0651